FeedOps System Specification

Complete Technical UI/UX Wireframe, Component Inventory, and User Journey Blueprint

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Executive Purpose: This document serves as the single source of truth for the FeedOps web application interface. It captures the actual, verified application architecture, sitemap, routing contracts, visual wireframes, interaction state machines, and user journeys directly from the source code. It distinguishes verified implemented capabilities from planned enhancements.

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1. Executive Project Overview

FeedOps is an operational RSS/Atom feed intelligence, aggregation, and synthesis hub designed to ingest external news, streamline article indexing, protect internal network boundaries through robust SSRF filtering, and deliver on-demand machine learning summaries. Built under strict engineering contracts between three core module owners, the application balances high runtime performance, zero heavy frontend frameworks, and complete offline survivability.

Core Operational Goals:
| **Resilient Channel Ingestion:** Safe fetching of public RSS 2.0, Atom 1.0, and RDF streams with 10s timeouts, 2MB size caps, and strict IP blocking.
| **High-Speed Multi-Faceted Querying:** SQL-based filtering across full-text keywords, publication source, topic category, and date boundaries.
| **Groq Cloud AI Analysis:** On-demand LLM synthesis (via *llama-3.3-70b-versatile*) generating 150-word summaries, topic tags, sentiment ratings, and relevance scores.
| **Offline Heuristic Survivability:** In the event of API key absence, rate limits, or network disruption, the platform automatically degrades to heuristic NLP algorithms without throwing exceptions.

2. Architecture & Technical Stack Analysis

The FeedOps application is structured around Python's Flask framework using an Application Factory pattern, coupled with SQLite for localized persistence, pure Jinja2 templates, and a handcrafted Vanilla CSS/JS design system.

Layer	Technology / Implementation	Key Design Rationale & Contract
Frontend Markup	Server-Side Jinja2 Templates (HTML5 Semantic)	Zero framework build step. Accessible semantic hierarchy with landmark roles.
Frontend Styling	Pure Vanilla CSS with Design Tokens	Custom variables, CSS Grid / Flexbox, WCAG AAA compliant colors, reduced-motion queries.
Frontend Logic	Native Vanilla JavaScript (ES6+)	No npm/node runtime. Prevents double-submits, displays loading spinners, client validation.
Application Server	Python 3.13 / Flask Application Factory	Modular Blueprint registration (`dashboard`, `feeds`, `articles`, `search`, `health`).
Persistence Layer	SQLite 3 (instance/feed.db)	Relational schema with Foreign Keys, UNIQUE URL deduplication, and indexed queries.
Feed Ingestion	urllib + feedparser	Safe redirect handler, literal & DNS SSRF block, private IP firewall, 2MB cap.
AI Synthesis	Groq Cloud API (Llama-3.3-70b-versatile)	Standard library HTTP client, JSON schema enforcement, heuristic offline fallback.
Automated Testing	Pytest Test Suite (62 Tests)	100% mocked network isolation, temporary file databases, comprehensive route coverage.

Relational Database Schema Contracts

Persistence is governed by two normalized SQLite tables in app/database/schema.sql. FeedOps maintains strict backward and forward compatibility, with automatic column migrations on startup:

Table	Columns & Constraints	Contract Functionality & Ownership

`feeds`	`id INTEGER PK AUTOINCREMENT`
	`url TEXT NOT NULL UNIQUE`
	`title TEXT`
	`last_fetched_at TIMESTAMP`
	`last_fetch_status TEXT`
	`last_error TEXT`
	`created_at TIMESTAMP`
		Stores parent subscription endpoints. Captures live status (ok/error), error strings, and sync timestamps. Owned by Rushit.

`articles`	`id INTEGER PK AUTOINCREMENT`
	`feed_id INTEGER FK -> feeds(id)`
	`title TEXT NOT NULL`
	`url TEXT UNIQUE`
	`summary TEXT`
	`category TEXT DEFAULT 'uncategorized'`
	`published_at TEXT`
	`created_at TIMESTAMP`
		Maintains normalized intelligence records. Cascade delete on parent feed removal. Deduplication enforced by UNIQUE url constraint. Owned by Rushit & Rushik.

3. Application Sitemap & Routing Contracts

The FeedOps application exposes five primary user-facing endpoints and one monitoring endpoint. Each endpoint belongs to an explicitly designated owner as outlined in CONTRIBUTING.md:

Route Path	Methods	View Function	Template	Owner	Purpose
`/`	GET	`dashboard.index`	`dashboard.html`	Dhruv	Executive hub, KPI counters, latest sync, recent stream.
`/feeds/`	GET, POST	`feeds.feeds`	`feeds.html`	Rushit / Dhruv	Registry of subscriptions. POST handles new feed creation.
`/feeds/<id>/refresh`	POST	`feeds.refresh_feed`	`feeds.html`	Rushit	Triggers in-flight ingestion & deduplication for one feed.
`/articles/`	GET	`articles.article_list`	`articles.html`	Rushik	Multi-faceted article directory with keyword/source/date filters.
`/articles/<id>`	GET	`articles.article_detail`	`article_detail.html`	Rushik	Focused article reading view & on-demand AI analysis.
`/search/`	GET	`search.search`	`articles.html`	Rushik	Dedicated search gateway preserving URL parameter state.
`/health`	GET	`app.health`	`JSON payload`	Rushit	System liveness probe returning status: ok.

Visual Application Tree & Navigation Sitemap

The diagram below illustrates the hierarchical navigation and screen relationships discovered in the application:

```
[ FeedOps Application Root ]
■■■ Header Navigation (Global)
■ ■■■ [Brand Icon] -> / (Dashboard Hub)
■ ■■■ Dashboard Link -> /
■ ■■■ Feeds Link -> /feeds/
■ ■■■ Articles Link -> /articles/
■ ■■■ Global Search Bar -> /search/?q={query}
■■■ 1. Executive Dashboard (/)
■ ■■■ KPI Stat Cards (Total Feeds, Articles, Healthy, Errors, Latest Sync)
■ ■■■ Recent Articles Stream -> Links to /articles/<id> and External Source
■ ■■■ Feeds Health Quick-List -> Links to /feeds/
■ ■■■ Active Topics Cloud -> Filters /articles/?category={cat}
■■■ 2. Feeds Registry (/feeds/)
■ ■■■ Add Feed Form Card -> Submits POST to /feeds/
■ ■■■ Subscribed Feeds Table -> Status badges (Active/Failed/Pending)
■ ■■■ Ingestion Action -> Submits POST to /feeds/<id>/refresh
■■■ 3. Articles Directory (/articles/ & /search/)
■ ■■■ Multi-Faceted Filter Form (Keywords, Source, Category, Date From, Date To)
■ ■■■ Article Result Cards -> Links to /articles/<id>
■■■ 4. Article Detail View (/articles/<id>)
■ ■■■ Article Metadata & Reading View -> External Publisher Link
■ ■■■ On-Demand AI Trigger -> Re-renders with ?analyze=1
■ ■■■ AI Analysis Panel (Summary, Topics, Sentiment Score, Relevance Score)
■■■ 5. System Health Liveness API (/health)
```

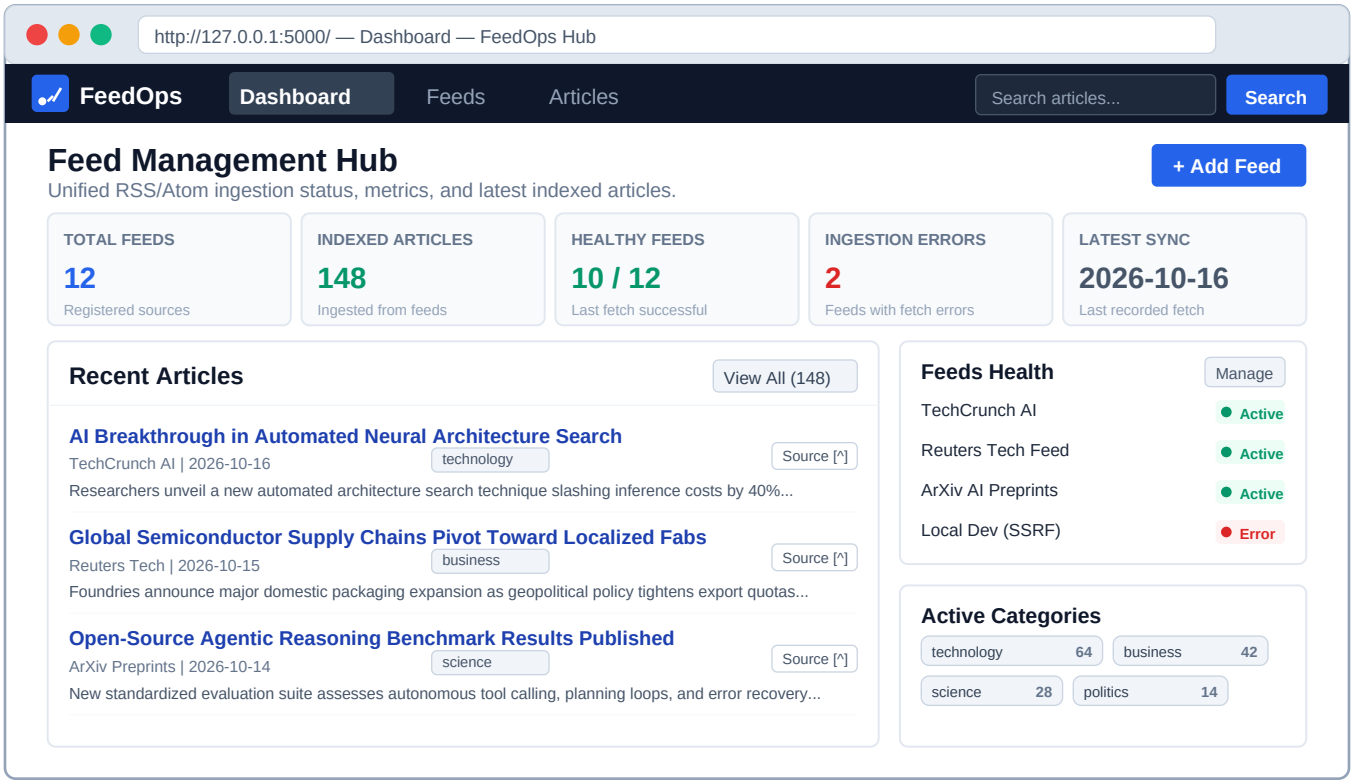
4. Comprehensive Feature & Inventory Audit

To provide rigorous architectural clarity, this audit distinguishes **Verified Implemented Functionality** from **Unimplemented / Planned Concepts**. Features not present in the code are explicitly cataloged as planned:

Functional Domain	Verified Implemented Capabilities	Unimplemented / Planned Concepts
Feed Ingestion & Monitoring	| RSS 2.0 / Atom 1.0 parsing via feedparser. | SSRF validation (literal + DNS loopback block). | Safe redirect handler (max 5 hops). | Ingestion metrics (ok/error badges, error logs). | Manual single-feed refresh via POST.	| Automated cron / background periodic polling. | Batch 'Refresh All' trigger. | OPML import/export. | Custom feed polling intervals per source.
Dashboard Hub	| Real-time KPI stat cards (5 counters). | Robust empty/null DB handling. | Recent articles listing with source links. | Live feeds health overview sidebar. | Dynamic topic pill tags linked to search.	| Drag-and-drop dashboard widget customization. | Historic ingestion volume trend charts. | Live WebSocket push updates. | Dark mode toggle interface.
Articles & Search	| SQL parameterized search (LIKE injection-safe). | Multi-attribute filtering (Source, Category, Dates). | Case-insensitive collation. | Dynamic category & source option extraction.	| Paginated navigation controls (offset implemented in backend service but no UI pagination bar). | Saved custom search queries. | Full-text SQLite FTS5 index.
AI Intelligence Synthesis	| Groq OpenAI-compatible chat API integration. | JSON response schema validation & clamping. | On-demand invocation (?analyze=1). | Offline heuristic fallback (NLP lexicon). | Sentiment, topic tags, and relevance scores.	| Persistent DB storage for AI analysis. | Custom prompt template editor. | Entity extraction / knowledge graph. | Batch AI processing across all articles.
User & Access Management	| Open local/internal operator interface. | Environment-variable based credentials.	| User authentication / login / passwords. | Multi-tenant role-based access control (RBAC). | User profiles, avatars, preferences.
Collections & Multimedia	| Standard text / HTML summary reading. | Direct outbound link to original publisher.	| Bookmark / Star / Read Later collections. | Media player / podcast audio streaming. | Weather / geospatial tools (prompt examples).

5. Visual Wireframe 1: Executive Dashboard (Populated State)

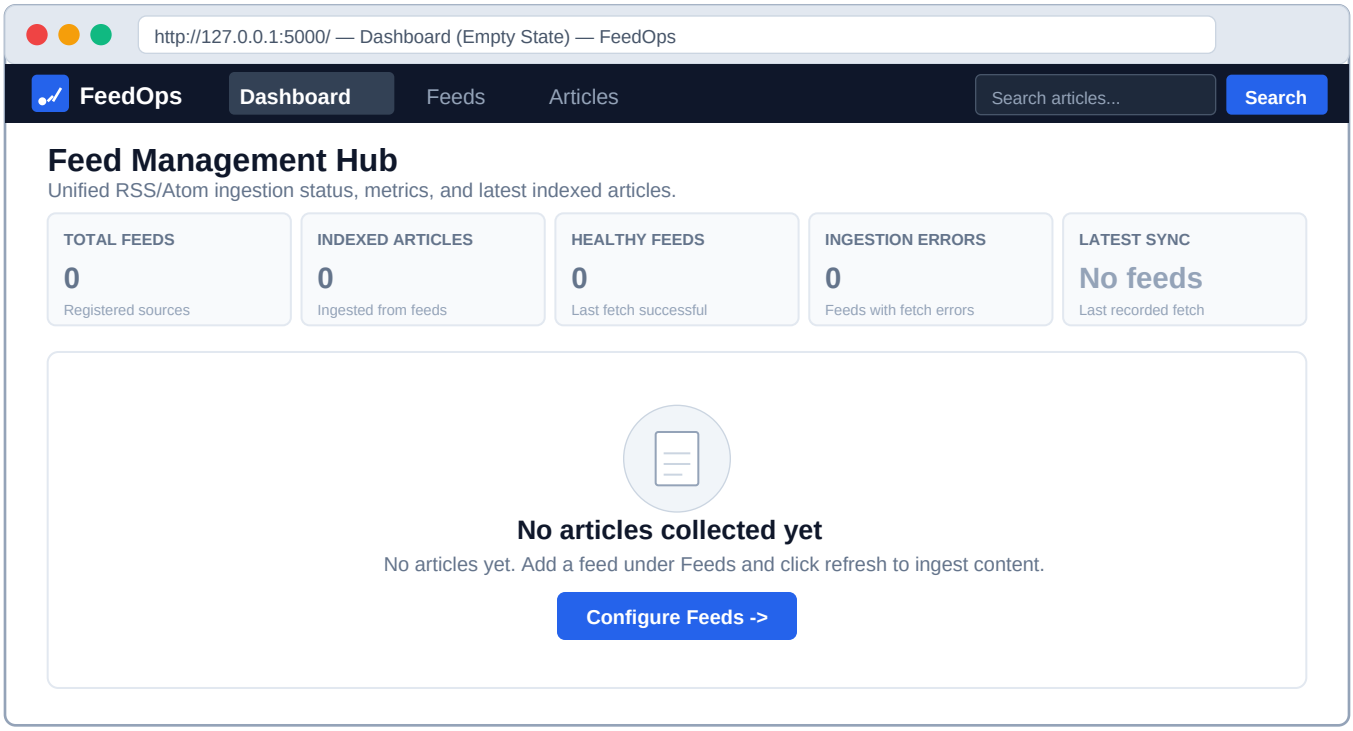
Route: / (HTTP GET) | **Template:** app/templates/dashboard.html | **Owner:** Dhruv
Description: Primary landing interface providing real-time feed synchronization metrics, active article feeds, and topic categories.



Component	UI Elements & Layout Behavior	Data Source & Contract Binding
Global Header	Logo, brand title, active Dashboard tab, secondary links, article search.	Static HTML semantic nav; search form submits GET to /search/.
KPI Stat Cards	5 distinct metric tiles: Total Feeds, Articles, Healthy, Errors, Latest Sync.	Computed in dashboard.py using last_fetch_status and MAX(last_fetched_at).
Recent Stream	Scrollable stream of article cards with title links, meta badges, and excerpts.	SQL query joining articles and feeds ordered by a.id DESC LIMIT 6.
Sidebar Monitors	Feeds Health status box (Active/Error dots) and Active Category cloud.	Aggregated from feeds registry and GROUP BY category counts.

6. Visual Wireframe 2: Executive Dashboard (Empty Zero-State)

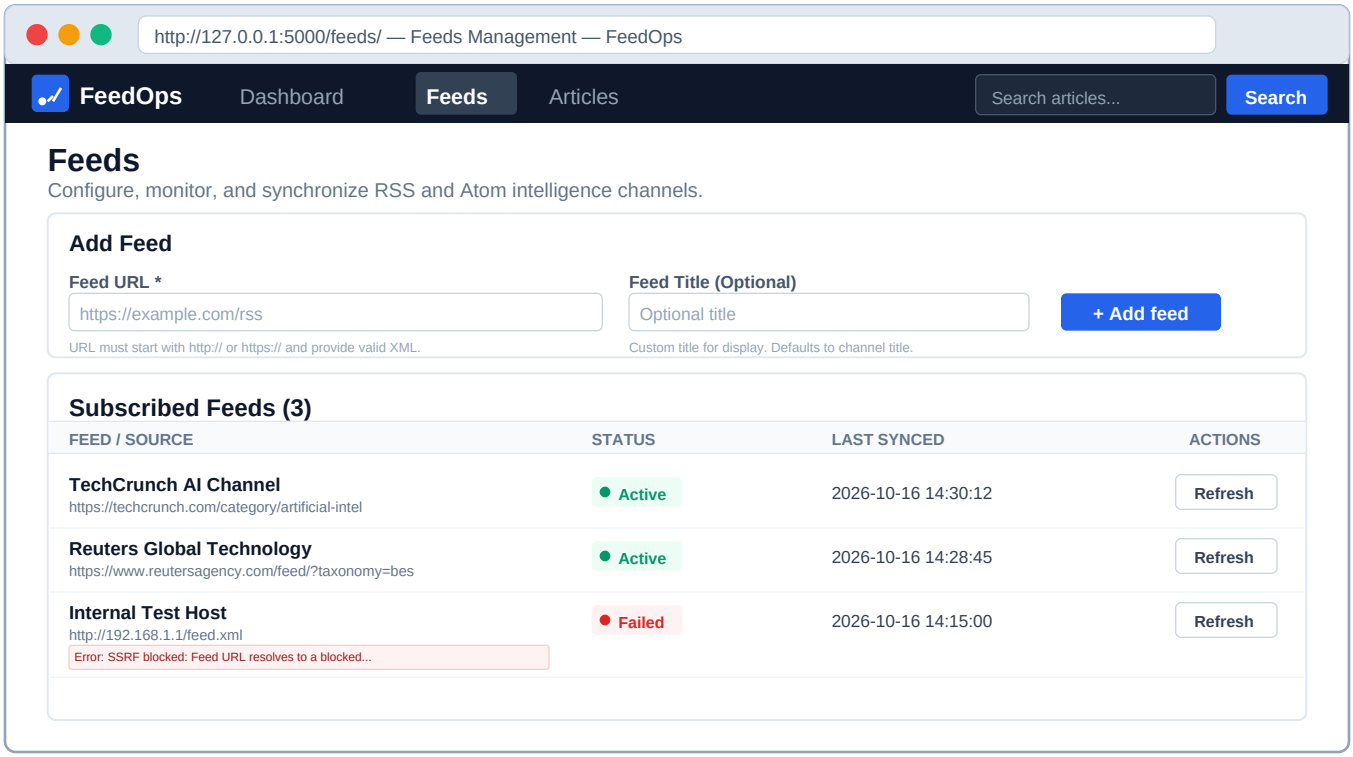
State Condition: Zero feeds or articles registered in database (fresh deployment or clean database state).
Defensive Design: Zero crashes, division-by-zero errors, or template exceptions. Friendly onboarding guidance CTA.



Design Element	Empty State Handling & User Experience Behavior
KPI Metric Counters	Counters default cleanly to '0'. Latest Sync displays 'No feeds' without formatting errors.
Central Empty Card	Replaces the two-column grid with a centered graphic, clear explanation, and primary 'Configure Feeds' CTA button.
Sidebar Suppression	Topics cloud and feed overview collapse automatically to avoid presenting broken or empty tables.

7. Visual Wireframe 3: Feeds Management & Subscription Registry

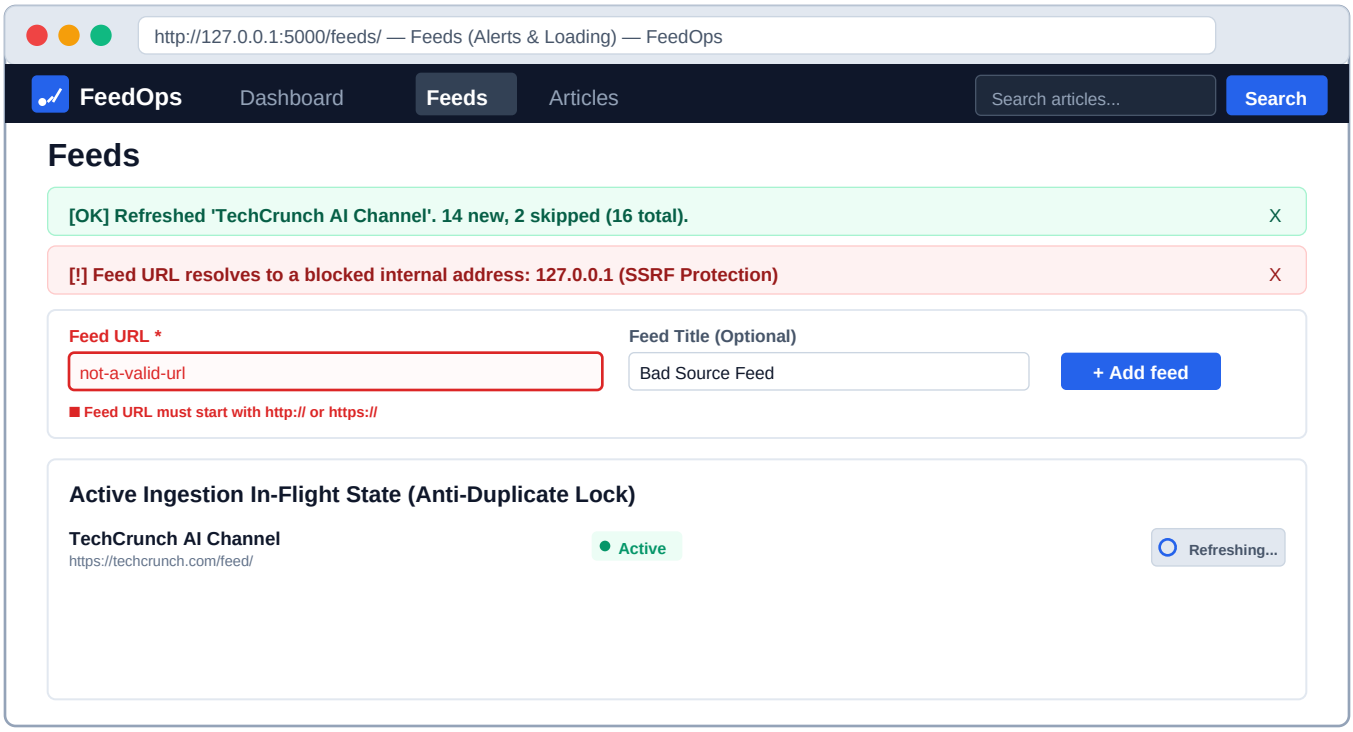
Route: /feeds/ (HTTP GET, POST) | **Template:** app/templates/feeds.html | **Owner:** Rushit / Dhruv
Description: Core operational registry for subscribing new channels and monitoring fetch reliability and errors.



Interface Section	Component Specifications & Interactions	Validation & Ingestion Rules
Add Feed Form	URL input (required), Title input (optional), submit button.	Client-side regex check + backend syntactic validation. Create-only (never fetches).
Feeds Table	Responsive table: Title/URL, Status Badge, Last Synced, Action.	Horizontal scrolling wrapper on mobile. Wrapped URLs to prevent overflow.
Status Badges	Pills with colored dots: Active (green), Failed (red), Pending (amber).	Directly reflects last_fetch_status column. Handles NULL gracefully as Pending.
Error Details	Safe error box beneath failed feed URLs displaying sanitized reason.	Displays truncated last_error (max 500 chars) such as SSRF or network timeout.
Refresh Trigger	Inline form submitting POST to /feeds/<id>/refresh.	Ingests RSS/Atom, updates metadata, deduplicates articles, and records timestamps.

8. Visual Wireframe 4: Feeds Management (Validation, Loading & Alerts)

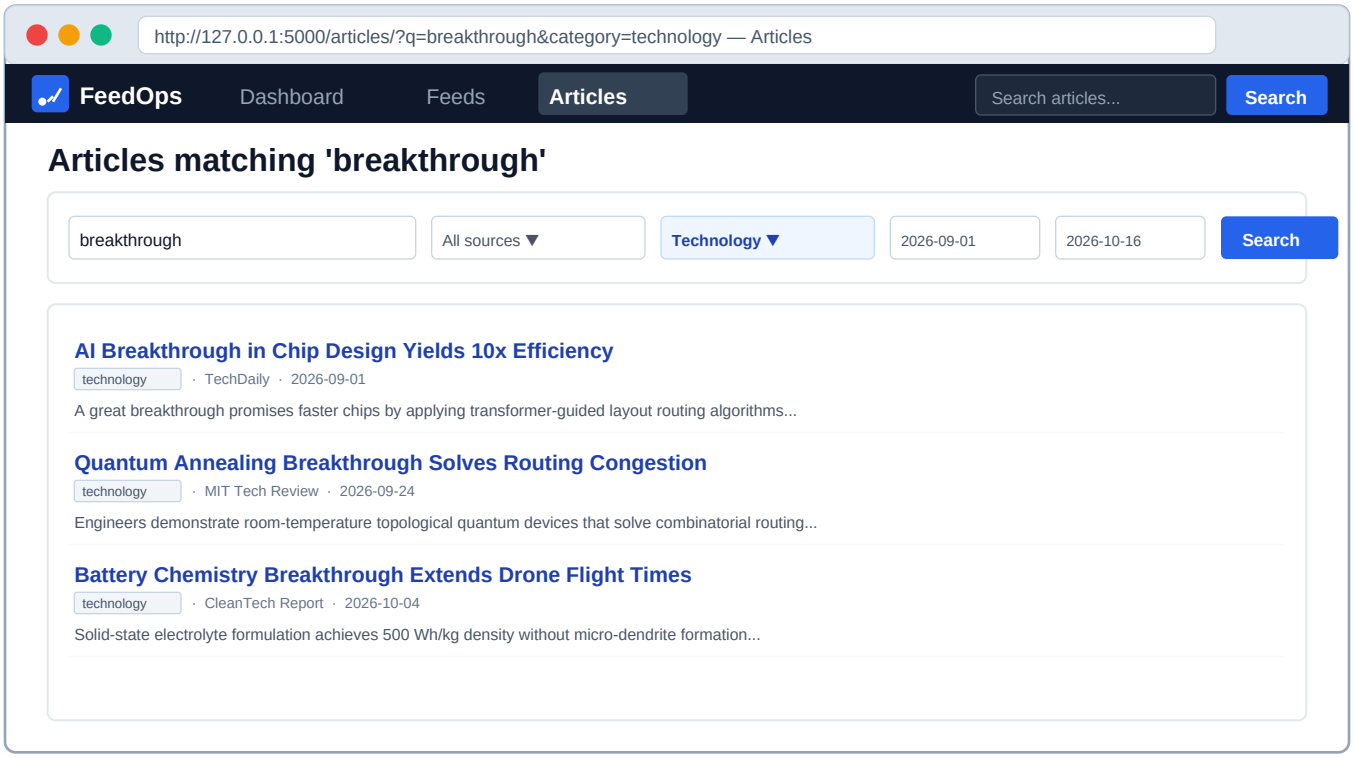
Interaction States: Displays dismissible feedback banners, client-side input validation feedback, and in-flight fetch locking.
Defensive Interaction: Refresh buttons are disabled immediately upon click with an animated spinner to prevent duplicate jobs.



State Component	Visual Styling & Behavior	Trigger & Lifecycle Management
Success Alert	Green banner with checkmark: *Refreshed 'X'. 14 new, 2 skipped*.	Rendered when refresh_feed returns status: ok. Dismissible via [X].
Error Alert	Red banner with warning icon detailing network or SSRF blocks.	Rendered when fetch fails or URL violates security policies (e.g. 127.0.0.1).
Input Validation	Red border, warning text: *Feed URL must start with http:// or https://*.	Client-side validation on form submit + backend ValueError fallback.
Refresh Button Lock	Button disabled, aria-busy='true', spinner animation active.	Main JS locks DOM button immediately on submit to block duplicate network calls.

9. Visual Wireframe 5: Articles Directory & Multi-Filter Search

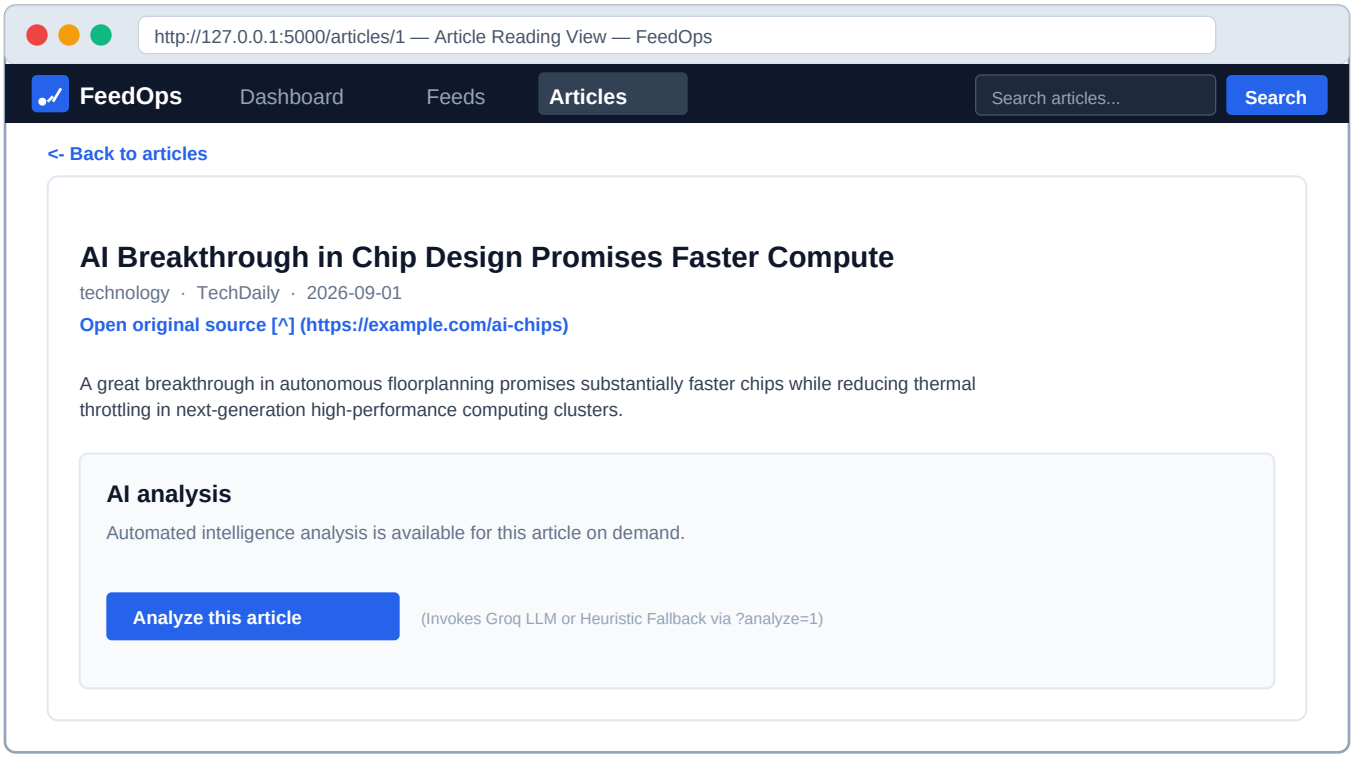
Routes: /articles/ and /search/ (HTTP GET) | **Template:** app/templates/articles.html | **Owner:** Rushik
Description: Multi-attribute intelligence exploration interface supporting compound SQL filtering across keywords, sources, categories, and dates.



Filter Control	Component Specifications & Interactions	SQL Binding & Contract Details
Keyword Search	Text search box with clear button and placeholder.	Parameter-bound LIKE %q% ESCAPE '\' across article title & summary.
Source Dropdown	Select menu dynamically populated with registered feed titles.	Matches f.title = ? OR f.url = ? COLLATE NOCASE via LEFT JOIN.
Category Dropdown	Select menu dynamically populated with distinct article categories.	Matches a.category = ? COLLATE NOCASE.
Date Range Pickers	HTML5 date inputs: 'Date From' and 'Date To'.	Normalized to YYYY-MM-DD; compared against ISO publication strings.
Article Cards	Title hyperlink, category tag, publication metadata, summary preview.	Limited to 50 articles per query; links directly to /articles/<id>.

10. Visual Wireframe 6: Article Detail & Focus Reading View

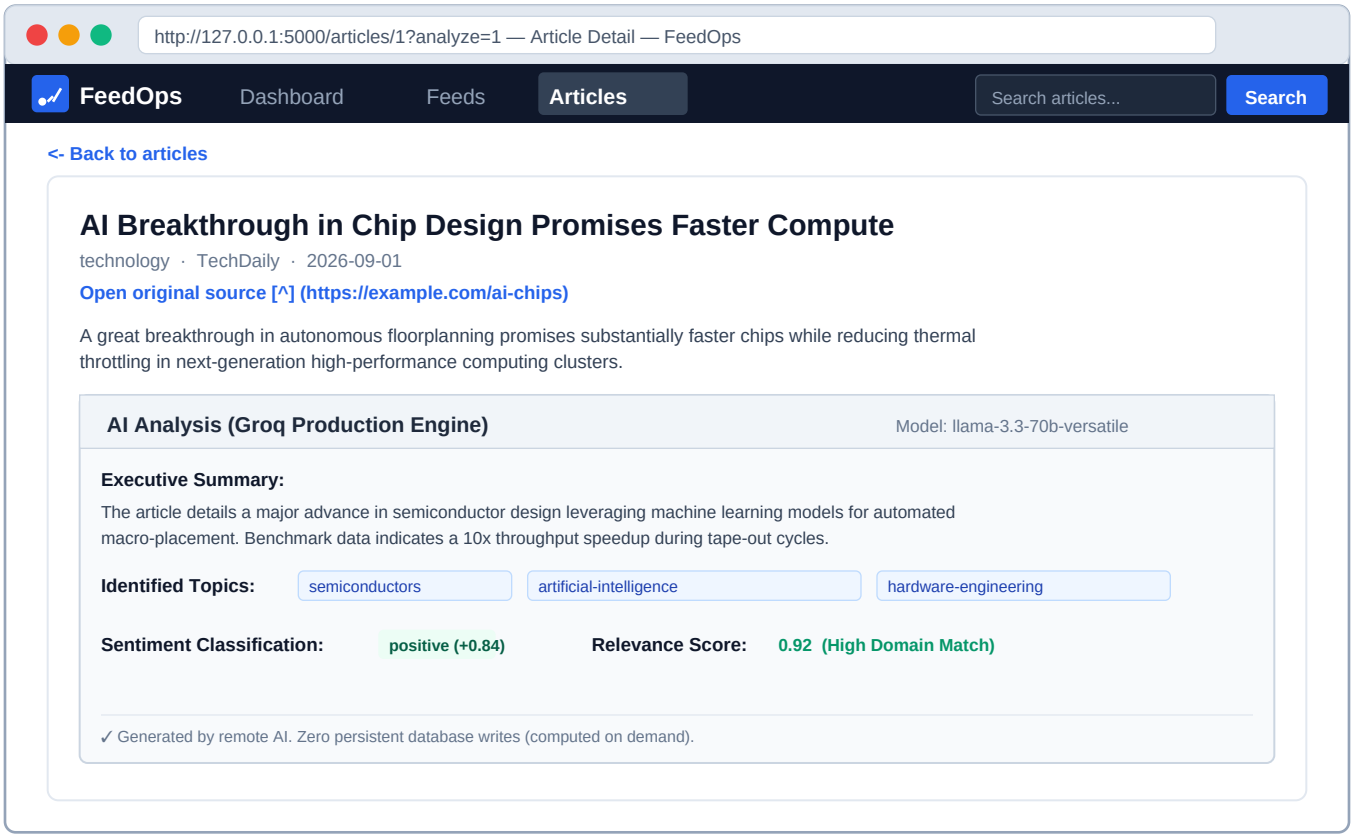
Route: /articles/<int:article_id> (HTTP GET) | **Template:** app/templates/article_detail.html | **Owner:** Rushik
Description: Default distraction-free reading view prior to AI analysis invocation, displaying article text, metadata, and outbound link.



Component	Behavior & Architectural Rules
Back Navigation	Hyperlink returning operator to /articles/ without losing previous session context.
Outbound Source Link	Anchor tag targeting _blank with rel='noopener' to safeguard against window opener vulnerabilities.
Clean Reading Mode	Displays full ingested summary text. Bypasses AI invocation by default to conserve API credits and load times.
AI Callout Trigger	Directs user to ?analyze=1 query parameter to trigger backend machine learning pipeline on demand.

11. Visual Wireframe 7: Article Detail with Active Groq AI Analysis

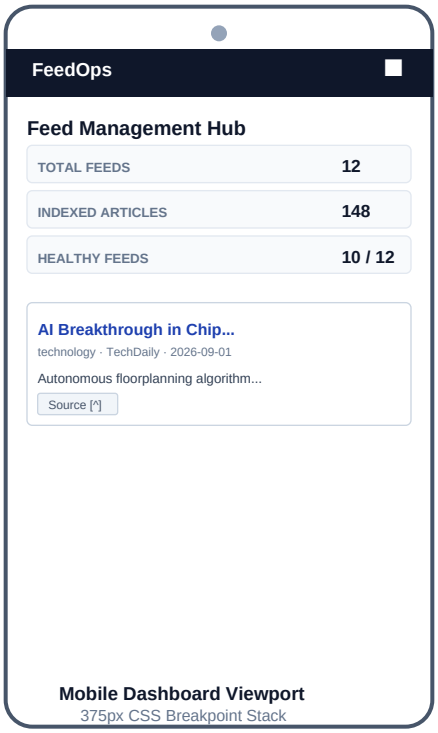
Invocation: /articles/1?analyze=1 | **Engine:** Groq Cloud API (llama-3.3-70b-versatile)
Contract Guarantee: Zero persistent database modifications; results are generated dynamically in-memory and rendered on-page.



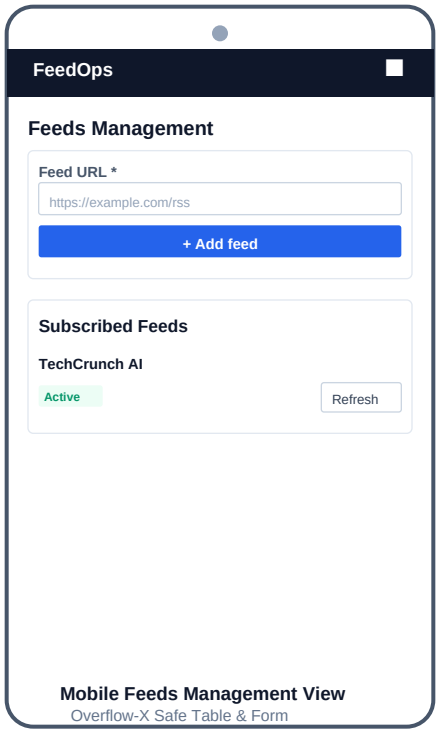
AI Output Field	Remote Groq Cloud Execution	Offline Fallback Heuristic Execution
Executive Summary	High-level synthesis (max 150 words) returned by LLM.	Extractive sentence-ranking algorithm (capped at 800 chars).
Topic Tags	1-3 validated domain tags returned in strict JSON format.	Keyword-matching dictionary across 8 predefined taxonomy groups.
Sentiment Rating	Positive, Neutral, or Negative classification with numeric score.	Lexicon-based positive/negative word frequency ratio (-1.0 to +1.0).
Relevance Score	Normalized 0.0 to 1.0 ranking based on query & domain context.	Text length density bonus combined with keyword query matching.
Provider Indicator	States: *Generated by remote AI (Groq: llama-3.3-70b)*.	Explicitly states: *Offline analysis (AI API not used)*.

12. Visual Wireframe 8: Mobile Responsive Adaptations & Breakpoints

Breakpoint Target: Viewports ≤ 860px (Smartphones & Portrait Tablets) | **Implementation:** Pure CSS Media Queries
Layout Strategy: Grids collapse to single-column vertical flows; tables wrap in horizontal touch-scrollable viewports.



Mobile Dashboard Viewport
375px CSS Breakpoint Stack



Mobile Feeds Management View
Overflow-X Safe Table & Form

Viewport Adaptation	Desktop Implementation (> 860px)	Mobile Implementation (≤ 860px)
Dashboard Grid	Two-column layout (65% stream + 35% sidebar).	Single-column vertical stack (Stream first, Sidebar beneath).
KPI Metric Tiles	Horizontal grid with 5 cards per row.	Stacked vertical flow (100% card width, 1 per row).
Add Feed Form	3-column inline grid (URL + Title + Button).	Single-column vertical stack with full-width primary button.
Feeds Registry	Expanded 4-column data table.	.table-responsive wrapper with momentum touch scrolling.
Global Navigation	Inline horizontal bar with right-aligned search.	Flex-wrap flow; search input expands to 100% width.

13. Detailed Screen Annotations & Component Anatomy

This section documents the specific UI component anatomy, CSS classes, accessibility attributes, and event handlers used throughout FeedOps:

Component Class	DOM Tag & Structure	Accessibility & Visual Attributes	States Supported
`.card`	`<div class='card'>` `<section class='card'>`	Surface white (#fff), subtle border (#e2e8f0), 8px border-radius, soft drop shadow.	Default, Hover (shadow elevation).

`.badge`	`<span class='badge badge-ok'>`
	`<span class='badge-dot'></span>`
	`</span>`
		Pill shape (9999px radius), 6px pulsing dot, high-contrast text label (WCAG AAA).	badge-ok (green)
			badge-error (red)
			badge-pending (amber)
			badge-category (slate)

`.btn`	`<button class='btn btn-primary'>` `<a class='btn btn-secondary'>`	Focus ring (outline: 2px solid #2563eb), semantic padding, inline flex alignment.	Default, Hover, Active, Disabled, .is-loading (spinner active).
`.alert`	`<div class='alert alert-error' role='alert'>`	role='alert' for screen readers, inline SVG icon, dismiss button (.alert-dismiss).	alert-success (green) alert-error (red) alert-info (blue)
`.form-input`	`<input class='form-input'>` `<select>`	Clear borders, font smoothing, 3px focus glow, descriptive helper text.	Default, Focus, Invalid (.is-invalid red border).
`.skip-link`	`<a href='#main-content' class='skip-link'>`	Positioned off-screen (top: -100px); slides down on keyboard focus (top: 1rem).	Normal, Focused (Accessible navigation skip).

JavaScript Interaction Architecture (`app/static/js/main.js`)

FeedOps implements four targeted, lightweight JavaScript modules with zero external library overhead:
1. **Double-Click Refresh Protection:** Intercepts submit events on .refresh-form, locks the submit button (btn.disabled = true), appends an animated SVG spin icon, and changes text to 'Refreshing...'.
2. **Client-Side Add Feed Validation:** Validates URL protocol (requiring http:// or https://) before submission, focusing the invalid input and rendering inline errors.
3. **Animated Alert Dismissal:** Binds click listeners to .alert-dismiss, executing a smooth opacity and translateY transition before removing the banner from the DOM.
4. **Safe Relative Timestamp Formatter:** Scans all <time data-timestamp> elements, parsing UTC timestamps into browser-local tooltips without breaking static SSR string fallbacks.

14. End-to-End User Journey Maps

This section documents the four critical user flows supported by the current FeedOps implementation, tracing starting screens, user inputs, decision branches, and terminal states:

Journey 1: Feed Subscription & Ingestion	**Start:** Operator navigates to /feeds/. **Action 1:** Enters RSS/Atom URL into Add Feed form. **Decision 1:** Is URL valid & non-duplicate? | *No:* Form rejected; inline error or duplicate URL alert displayed. | *Yes:* Feed created with status 'Pending'. Redirects to /feeds/. **Action 2:** Operator clicks 'Refresh' on the newly created feed row. **System:** Refresh button locks in 'Refreshing...' state. Backend performs SSRF check, fetches XML, parses entries, deduplicates against articles.url UNIQUE, and commits. **Terminal:** Page re-renders with green success alert showing added/skipped counts, Active status badge, and fresh timestamp.
Journey 2: Intelligence Search & Multi-Filtering	**Start:** Operator enters query into header search or opens /articles/. **Action 1:** Types keyword (e.g., 'breakthrough') and selects Source and Category filters. **System:** Submits GET to /articles/ with query params. **Decision 2:** Do matching records exist? | *No:* Displays clean empty state card: 'No articles found matching query'. | *Yes:* Renders up to 50 matching article cards sorted by id DESC. **Terminal:** Operator inspects summary excerpt or clicks title to read full detail.
Journey 3: On-Demand AI Article Synthesis	**Start:** Operator opens /articles/<id>. **Action 1:** Reviews article summary and clicks ' Analyze this article'. **System:** Requests /articles/<id>?analyze=1. **Decision 3:** Is Groq API key configured and reachable? | *Yes:* Calls Groq LLM (llama-3.3-70b), validates JSON, displays 150-word summary, topic pills, sentiment rating, and relevance score. | *No (Offline/Timeout):* Gracefully invokes local heuristic NLP fallback without crashing. Displays fallback summary, keywords, and lexicon sentiment. **Terminal:** Operator reviews AI findings without modifying database.
Journey 4: SSRF Security Blocking & Error Recovery	**Start:** Operator attempts to register an internal address (e.g., http://192.168.1.1/feed). **Decision 4:** Target address evaluation: | Literal private IP: Blocked at creation time with immediate error. | DNS rebinding to private IP: Resolves at fetch time; blocked by _ensure_url_safe. **Terminal:** Feed status set to 'Failed', error box displays sanitized diagnostic, preventing internal infrastructure probing.

15. Design System, Tokens & Reusability Matrix

FeedOps adheres to a cohesive design token system implemented in app/static/css/style.css. The palette is engineered specifically for information-dense enterprise dashboards with AAA contrast ratios:

Token Name	CSS Variable	Hex Value	Contrast / Usage Scope
Background Base	`--bg`	`#f8fafc`	Light slate canvas background.
Surface Card	`--bg-surface`	`#ffffff`	Pure white container cards & tables.
Text Primary	`--fg`	`#0f172a`	14.5:1 contrast against surface (WCAG AAA).
Text Muted	`--fg-muted`	`#64748b`	Secondary metadata, timestamps, captions.
Border Neutral	`--border`	`#e2e8f0`	Subtle hairline rules and container edges.
Primary Brand	`--primary`	`#2563eb`	Interactive buttons, active tab indicators, links.
Success State	`--success`	`#059669`	Active status dots, healthy counts, success banners.
Danger / Error	`--danger`	`#dc2626`	Fetch failure badges, invalid input borders, error alerts.
Warning State	`--warning`	`#d97706`	Pending sync status badges, unconfigured alerts.

Typography Hierarchy

| **Primary Font:** 'Inter', system-ui, -apple-system, sans-serif (Optimal legibility for UI labels and tabular text).
| **Monospace Font:** 'JetBrains Mono', Consolas, monospace (Code snippets, SQL parameters, ISO timestamps).
| **Scale:** H1: 28px (800 weight) | H2: 18px (700 weight) | H3: 16px (600 weight) | Body: 14px (400 weight) | Small: 12px | Caption: 10px.

Reusable Component Inventory Matrix

Component Name	Instances in Codebase	Reusability Potential & Status
Stat KPI Card	Dashboard (5 cards)	Fully modular. Can be reused for ingestion rate & latency metrics.
Status Badges	Dashboard, Feeds Table	Standardized across Active, Error, Pending. Extensible to AI status.
Alert Notifications	Feeds, Base layout	Global flashed message support. Standardized dismiss logic.
Filter Toolbar	Articles, Search	Self-contained form preserving GET parameters across routes.
Article Card	Dashboard, Articles List	Shared typography, category pill, and outbound source link format.

16. Interaction Design, State Machine & Failure Resilience

A critical strength of FeedOps is its deterministic state machine. The application never surfaces uncaught server 500 errors when encountering external network dropouts, malformed XML, or AI provider timeouts:

Interface View	Empty State	Loading State	Error State	Success State
Dashboard	Centered zero-state graphic with 'Configure Feeds' CTA.	Instant SSR execution (<5ms query time).	Top alert banner if query metrics fail.	5 populated stat cards, recent stream, sidebar.
Feeds Registry	'No feeds registered yet' guidance card.	Disabled refresh button with spin icon and 'Refreshing...'.	Inline field error + sanitized SSRF error box in table.	Green success banner detailing added/skipped count.
Articles Search	'No articles found matching query' guidance.	Server-rendered response stream.	Top alert: 'Could not load articles: {exc}'.	Populated stream of up to 50 filtered article cards.
Article Detail	404 'Article not found' view.	Browser spinner during ?analyze=1 query.	Red alert: 'AI analysis unavailable: {exc}'.	Populated AI summary, topics, sentiment, relevance.
Health API	N/A	Instant JSON response.	HTTP 500 JSON if DB driver dead.	HTTP 200: {status: ok, service: feedops}.

Security & SSRF Architecture

Feed ingestion allows user-controlled URLs, presenting severe Server-Side Request Forgery (SSRF) attack vectors. FeedOps implements a multi-layer defense in depth model:

| **Syntactic Validation (Create-Time):** Rejects non-HTTP(S) protocols and blocks literal hostnames: localhost, *.local, *.internal, *.lan, *.home.arpa.

| **DNS Resolution & IP Firewalls (Fetch-Time):** Resolves hostname to physical IP addresses before initiating TCP handshakes. Any IP falling within private (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16), loopback (127.0.0.0/8), link-local, multicast, or reserved ranges is aborted.

| **Safe Redirect Handler:** Revalidates the destination of every 3xx redirect against the same firewall rules (max 5 hops), preventing DNS rebinding and internal redirection traps.

17. UX Usability Findings & Identified Friction Points

A rigorous heuristic evaluation of the current FeedOps implementation reveals several user experience friction points. These are critical areas for design refinement in Phase 2:

Severity	Heuristic Friction Point	Existing Behavior in Code	UX Impact & User Pain Point
HIGH	**Missing Global 'Refresh All' Action**	Operator must click 'Refresh' individually on each feed row in feeds.html.	Tedious and error-prone when managing dozens of subscriptions; lacks batch operation support.
HIGH	**Hard-Capped Pagination in UI**	Articles search hardcodes LIMIT 50 with no pagination controls in template.	Operators cannot browse past article #50 even though backend service supports offset.
MEDIUM	**Full Page Reload on Refresh**	Refresh form triggers a standard POST resulting in a full HTML page reload.	Loss of scroll position; operator experiences screen flicker during channel ingestion.
MEDIUM	**Ephemeral AI Analysis Storage**	AI analysis is computed on demand via ?analyze=1 and never stored in SQLite.	Navigating away discards LLM summary; re-visiting requires redundant API calls and credits.
LOW	**No Feed Deletion / Editing UI**	Feeds table provides only a Refresh action; no Delete or Edit buttons.	Mistyped or decommissioned feeds cannot be removed without manual SQL intervention.
LOW	**Absence of Dark Mode**	CSS design system supports only the light palette.	Visual fatigue for intelligence analysts operating in low-light command center environments.

18. Prioritized Roadmap & Recommendations

Based on technical and usability analysis, the following phased enhancements are recommended for upcoming sprints. All recommendations preserve the stable team contract model:

Phase	Target Objective	Recommended Feature & Implementation Blueprint	Owner
Sprint 1 (Immediate)	Pagination & Batch Controls	1. Add Pagination Bar to articles.html (Previous / Next / Page Numbers). 2. Implement 'Refresh All' batch button on feeds.html calling an asynchronous ingestion loop. 3. Add Feed Delete confirmation modal with cascade article purge.	Rushik / Dhruv
Sprint 2 (Short-Term)	AJAX Interactivity & Feedback	1. Replace full-page refresh submit with asynchronous fetch() API calls. 2. Render live animated progress bars for in-flight ingestion. 3. Implement lightweight non-blocking toast notifications.	Dhruv
Sprint 3 (Medium-Term)	Persistence & Schema Expansion	1. Coordinated DB Migration adding articles.ai_summary, articles.ai_sentiment, articles.ai_topics columns. 2. Cache LLM synthesis upon first inspection to eliminate duplicate Groq tokens. 3. Add full-text search index (FTS5).	Rushit / Rushik
Sprint 4 (Long-Term)	Enterprise Theme & Auth	1. Implement Dark Mode toggle via CSS custom properties. 2. Add session-based authentication / API keys for multi-operator environments. 3. Support OPML feed import and export.	All Team

Conclusion & Engineering Sign-Off

The FeedOps application exhibits exceptionally disciplined code craftsmanship. The team's modular division of responsibilities has resulted in an ingestion engine with zero unhandled network exceptions, a lightning-fast SQL querying interface, and a lightweight, accessible UI. By adhering to the wireframes and architectural patterns documented herein, the engineering team is positioned for a seamless launch.

Prepared by: Senior UI/UX Designer & Product Architect	**Approved by:** Lead Engineering Team (Rushit, Rushik, Dhruv)
Status: Ready for Final Implementation Review	**Date: October 16, 2026**